Final Project: Advanced Machine Learning
Home Credit Default Risk
Predicting Repayment Abilities

Dr. Jaume Manero

May 2026

Abstract

This project challenges to broaden financial inclusion for the unbanked population. Using the Home Credit Default Risk dataset, students will leverage alternative data—including telco and transactional information—to predict client repayment abilities. This task requires navigating a complex relational database, handling significant class imbalance, and ensuring model interpretability in a regulated financial context. The ultimate goal is to reach the top 5% of the Kaggle leaderboard.

Contents

1 Project Overview and Business Context

Many people struggle to get loans due to insufficient or non-existent credit histories, often falling prey to untrustworthy lenders. Home Credit strives to provide a safe borrowing experience for the underserved. By unlocking the full potential of alternative data, we ensure that capable clients are not rejected and are provided with loan terms that empower their success.

The key of this competition is to solve a few issues in the data and understand the complexity of the problem of Credid Default Risk. I recommend to read the article on this subject [**?**] and to look at published material from this competition. Look for information on the 1st place competition "Home Aloan" which has a complex and effective feature engineering approach.

Additionally, look at two points used in the best solutions, which are:

- **The 365243 Replacement:** In the DAYS_EMPLOYED column, the value 365243 is a common anomaly that acts as a sentinel for "infinity" or "unknown." Top solutions replace this with NaN to prevent the model from treating it as a literal number of days.
- **One-Hot Encoding with NaNs:** Using dummy_na=True during one-hot encoding allows the model to learn if the absence of a specific data point (like a missing credit card limit) is a predictor of default risk.

1.1 The Challenge

The primary objective is to build a predictive model that identifies the probability of a client defaulting on a loan. Unlike previous exercises with single-table datasets, this project involves:

- **Relational Data:** Aggregating information across seven distinct tables.
- **Class Imbalance:** Managing a target variable where only 8% of clients default.
- **Explainability:** Providing "Force Plots" to justify credit decisions.

2 Data Architecture

The dataset is composed of several relational tables that must be joined and aggregated:

- **application_{train—test}.csv:** The main table containing static data for all applications. One row per loan.
- **bureau.csv:** All client's previous credits from other financial institutions reported to the Credit Bureau.
- **bureau_balance.csv:** Monthly balances of previous credits in the Credit Bureau.
- **POS_CASH_balance.csv:** Monthly balance snapshots of previous point-of-sales and cash loans with Home Credit.
- **credit_card_balance.csv:** Monthly balance snapshots of previous credit cards with Home Credit.

- **previous_application.csv:** All previous applications for Home Credit loans by clients in the sample.
- **installments_payments.csv:** Repayment history for previously disbursed credits.

3 Evaluation Metric: AUC-ROC

Due to the 8% default rate, accuracy is a misleading metric. We use the **Area Under the ROC Curve (AUC-ROC)**.

$$\text{TPR (Recall)} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \tag{1}$$

$$\text{AUC} = \int_0^1 \text{TPR}(t)\, d(\text{FPR}(t)) \tag{2}$$

An AUC of 1.0 represents a perfect classifier, while 0.5 represents random guessing. Students must optimize their models to rank the risk of applicants effectively.

4 Project Phases (The 7-Step Workflow)

Students must clearly mark these seven phases in their Jupyter Notebook and Project.

4.1 Phase 1: Dimensionality Reduction

The data is spread across 7 tables. You must aggregate these (using mean, max, min, sum) which will result in hundreds of features. Use **Principal Component Analysis (PCA)** or **Feature Importance** from a baseline model to prune noise.

4.2 Phase 2: Imbalance and Probabilistic Modeling

With only an 8% default rate, move beyond accuracy. Implement **Bayesian Networks** to model the behavioral reasons for default and evaluate using AUC-ROC or PR-Curves.

4.3 Phase 3: Time Series Lite (Temporal Engineering)

The `credit_card_balance` and `installments` tables are temporal. Create meaningful predictors using **lagging** (e.g., "What was the balance 3 months ago?" or "Trend in late payments").

4.4 Phase 4: Feature Engineering Final

Explore alternative Feature engineering approaches. Research and test different techniques.

4.5 Phase 5: Ensemble (Battle of the GBMs)

Compare the performance of **XGBoost** vs. **LightGBM**. Focus on how each handles the large volume of categorical variables and the high-dimensional feature space.

4.6 Phase 6: Submission Generation

Generate a CSV submission file containing the `SK_ID_CURR` and the predicted probability for the `TARGET` variable.

4.7 Phase 7: Interpretability

In banking, "black box" models are unacceptable. Use **SHAP (SHapley Additive exPlanations)** to generate a Force Plot for a specific rejected applicant to explain which features (e.g., debt-to-income ratio) triggered the rejection.

5 Submission and Competition

Kaggle Competition Link:
`https://www.kaggle.com/competitions/home-credit-default-risk/`

Target Goal

To achieve a passing grade for the competitive portion, your model should aim for a position within the **top 5%** of the Kaggle leaderboard. Do not overcomplicate; focus on robust feature engineering and clean cross-validation.

6 Presentation in Class

In Class 13/14 each group will present its project to the class. The presentation must communicate your learning journey and the results of your best project. A good storyline for your presentation could be:

- First Baseline of the project and result
- Research and ideas applied in the following versions
- Second Version and result
- Final approach
- Final result and position in the leaderboard
- Lessons Learned and what would you do next

7 Deliverables

Make sure all your material has the names of all your group memebers.

1. **Jupyter Notebook:** Clearly divided into the 6 Phases.
2. **Leaderboard Screenshot:** Evidence of your Kaggle score and ranking.
3. **Interpretability Report:** A brief analysis of the SHAP values for at least two extreme cases (one high risk, one low risk).
4. **Class presentation:** The material (powerpoint, canvas, ..) used in the class presentation